

7. [Maximum mark: 5]

At 09:00 a helicopter is located at a point $(10, 3, 0.5)$ relative to a point O on horizontal ground. The x -direction is due east, the y -direction is due north and the z -direction is vertically upwards.

All distances are measured in kilometres.

The helicopter is flying at a constant height.

The helicopter's position relative to the point O is given by $\mathbf{r} = \begin{pmatrix} 10 \\ 3 \\ 0.5 \end{pmatrix} + 4t \begin{pmatrix} 10 \\ -25 \\ 0 \end{pmatrix}$, where t represents the time in hours since 09:00.

(a) Find the speed of the helicopter. [2]

At 10:00 the helicopter begins to descend.

During descent the helicopter's vertical height decreases at a constant rate of 16 km h^{-1} and its horizontal velocity remains unchanged.

The angle of descent, β , is defined as the angle between the helicopter's direction of travel and the horizontal.

(b) Find β , giving your answer in degrees. [3]

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